

Linux Containers Explained

Linux containers (LXCs) perform better and faster than virtual machines. We take a quick look at why this is so, compare Docker with a Linux container, and then explore the advantages of LXCs.



Containers behave like virtual machines. However, unlike virtual machines, which must replicate an entire operating system to function, containers only need to replicate the specific components they require to function. This lowers the application's size and provides a noticeable performance boost. They also function far more quickly because, in contrast to traditional virtualisation, they essentially run on their host natively, albeit with an extra layer of security.

Linux containers started out as an open source project to offer an operating system-level virtualisation

technique that makes use of the built-in capabilities of the Linux kernel. With major support from IBM, among others, the project began in the late 2000s and soon found its way into the mainline Linux kernel. Through this integration, LXCs were able to take advantage of native support and optimisations, which aided in their uptake and continued improvement.

Multiple isolated Linux systems, or containers, can be run on a single Linux host using Linux containers. LXCs enable containers to function as standalone systems with their own file system, process tree, and network

interfaces while sharing the host kernel. They are frequently thought of as a more portable substitute for virtual machines and have paved the way for later advancements in containerisation, such as the creation of Docker.

Docker vs LXCs

In the Linux environment, containerisation technologies like Docker and LXC are similar, but they address different use cases and methods of deployment and isolation. Table 1 gives a quick comparison of Docker and LXC.